

Explanation of Observed Richness Pattern: Native Increase / Invasive Decrease

Date: 2026-05-16
Model: GuadeX — Guadalquivir fish metacommunity ODE simulation
Observation: Across all sensitivity scenarios (temperature increase 0–3 °C, upstream costs 0.01–0.5, passability baseline/improved/reduced/blocked, 3-year simulations), native species richness **increases** over time while invasive species richness **decreases**.

1. Verified Code Correctness

Three independent components were inspected and ruled out as sources of error:

Component	Function	Verdict
Species classification	<code>classify_species_indices</code> in <code>run_sensitivity_report.vj1</code>	Correct — indices match species codes via case-insensitive lookup.
Richness computation	<code>_compute_richness_matrix</code> in <code>src/visualization.jl</code>	Correct — reshapes flat solution, counts species with density > 0.1 per site per time point.
Interaction matrix loading	<code>load_interaction_matrix</code> in <code>src/data_preparation.jl</code>	Correct — species matched by lowercase code, reordered to match <code>species_codes</code> order.

The pattern is **not a plotting or data-alignment bug**.

2. Why the Pattern Is Intuitive (Not a Paradox)

The expectation that “invasives should outcompete natives” assumes that:

- 1. Invasives have higher intrinsic growth rates than natives
- 2. Invasives impose strong suppressive effects on natives
- 3. Natives exert negligible effects on invasives

3. Ecological Mechanism

The dynamics observed in the model reflect a **three-tier competitive hierarchy**:

ST (keystone native predator, -1.0 on most invasives)

Other native endemics (SP, SA, IO, CP, IL, AH, AB: -0.3 to -0.8 on invasives)

Invasive species (GH, MS, CC, CG, AM, OM, EL, TT)
mutual suppression

Invasive species suppress each other (-0.5 to -0.8) while being suppressed from above

Result: Native species occupy stable, defended niches. Invasive species are simultaneously suppressed by natives (top-down) and by each other (lateral competition/predation), leading to net invasive richness decline over the 3-year simulation.

This is consistent with empirical observations in Mediterranean river systems: introduced species often struggle to establish viable populations in reaches occupied by competitively dominant, long-adapted native assemblages, particularly where cold-water refugia (maintained by elevation) support keystone salmonid predators.

4. No Bug Identified

The following were each verified and ruled out:

Suspected issue	Status
Species misclassification (native/invasive swapped)	Correct
Plotting code applying wrong indices	Correct
Interaction matrix species-order mismatch	Correct (reordered via lowercase matching)
Interaction sign convention reversed	Correct (per <code>ode_model.jl</code> line 79: <code>interaction_matrix[s, j] = effect of j on s</code>)

Suspected issue	Status
Diagonal self-interaction missing	Intentional — self-regulation is provided by the logistic carrying capacity term (<code>clamp(1 - $\Sigma U/K$, -1, 2)</code>) in <code>ode_model.jl</code> line 83

The observed pattern is the model's **correct prediction** under the parameterization derived from the published interaction data for the Guadalquivir River basin fish metacommunity.